RESEARCH PROJECT TOPICS

TUTORIAL 1

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(lecture adapted from that of P.Isenberg, used with permission)

CONSTRAINTS

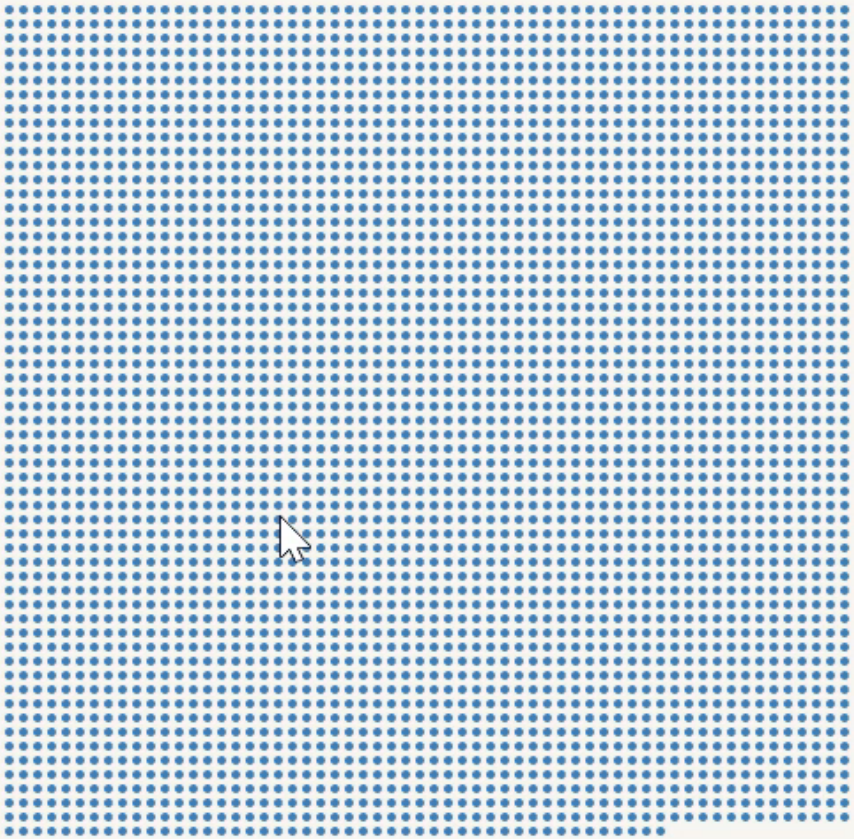
Work on a project related to **socio-economic issues**

Climate change, gender equality, fights against diseases, poverty, tech industry jobs…

Groups of 3

EXAMPLE CLASS PROJECTS

Perception of migration on social media - U.S. vs U.K



≈ 351000 tweets 2 million retweets 5 million likes

Each circle represents 100 tweets. Tweets were collected from Aug. to Oct. 2019

70.8 million people are forcibly displaced worldwide for reasons such as poverty or war.

The nearly **351,000 migration-related tweets** represented here as circles were collected during the months of August, September and October from the United States and the United Kingdom.

Social media is a rich source of information to observe and understand the views of the general population towards migrants and refugees: **how much the matter concerns them, how they feel about it and what events are perceived as particularly important**.

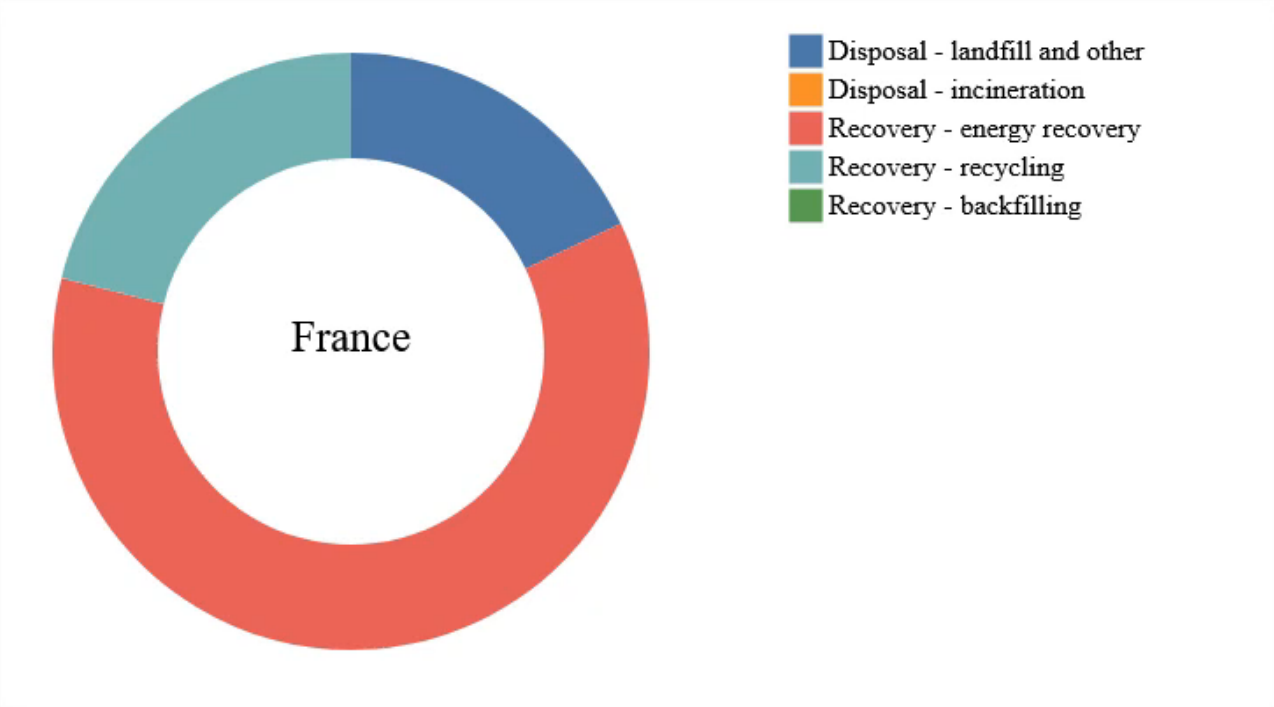
SCROLL DOWN

How is your country treating its plastic waste?

Although treating the total amount of plastic waste generated should be a priority, management operations have different impact on the environment. In particular, recovery (recycling, energy recovery and backfilling) is greener than disposal (incineration and disposal in landfills).

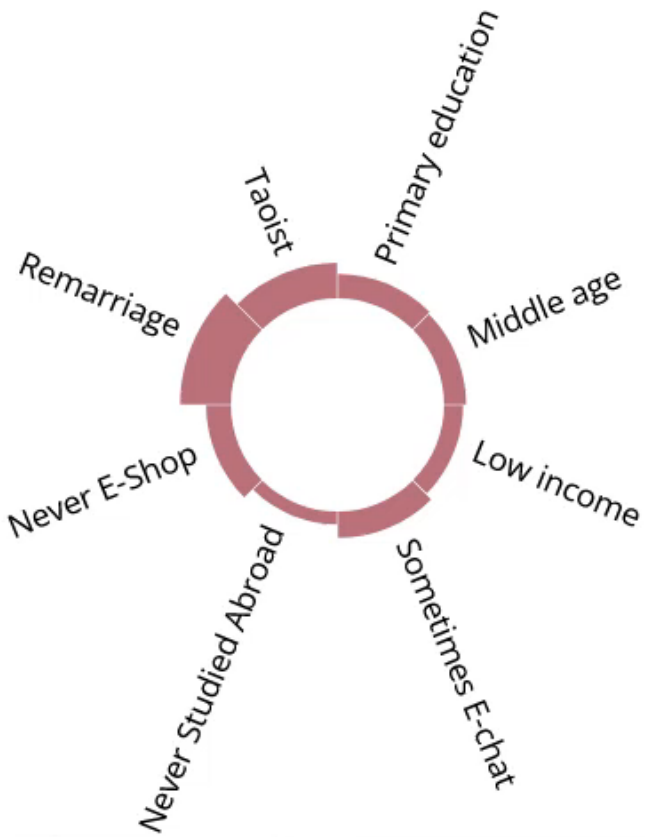
Select a country: France

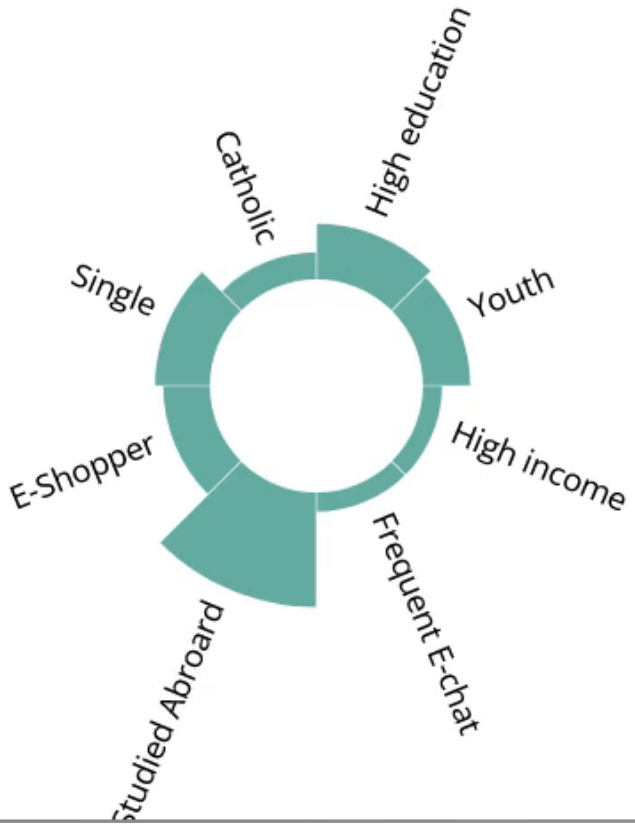
The different plastic waste management operations of each country are displayed in a **donut chart**. A country can by picked by choosing from the selector. The goal of this visualisation is to allow the user to check how a particular country treats its plastic waste.



Groups Of Different Categorical Attributes Towards LGBT

This circular bar chart means to represent the effect of **single categorical factor** on the attitudes towards LGBT.
We construct it based on the how much score higher/lower than global average when it belongs to certain category.





PROBLEMS WITH PAST PROJECTS

Topic was not of societal importance

“Blockchain’s Market Cap”

PROBLEMS WITH PAST PROJECTS

Good topic but boring research question

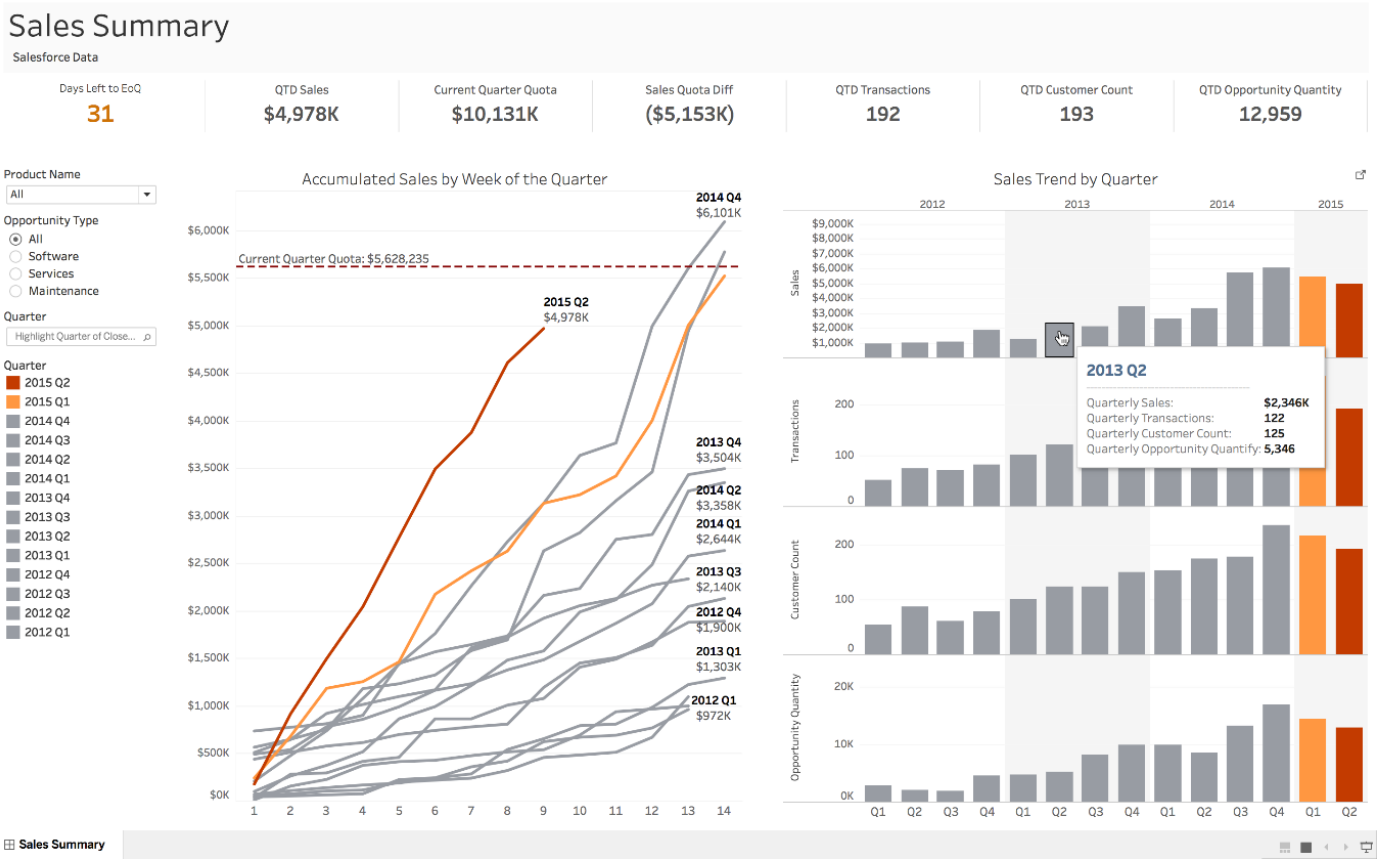
CHARITY AND DIFFERENT BEHAVIORAL PATTERNS OF PARTICIPATING IN IT → What country has the highest number of charity organizations?

PROBLEMS WITH PAST PROJECTS

Good topic and question but stakeholders too broad

Climate Change => visualization for everyone

WHAT NOT TO DO



Do not build a general purpose visualization tool

(Like Tableau/Excel/PowerBI...)

HOW TO FIND A PROJECT?

Two approaches

- Start from your data -> explore (data-driven research)
- Start from questions -> search for data (task/question-driven research)

If you have data, focus on them, else …

YOUR STAKEHOLDERS

Knowing who your project is for is important to narrow down your topic and to focus your research questions

E.g., Young adults with some level of higher education (university students)

FIND A TOPIC YOU ARE EXCITED ABOUT

1. Abortion
2. Age, Aging, Age Discrimination, Death
3. Behavior Modification
4. Charity (as a social issue)
5. Communication (Social issue)
6. Drug use (Social issue)
7. Economy and Poverty (Social Issue)
8. Food (Social issue)
9. Gender / Sexism / Feminism / Masculism (Social Topic)
10. Housing (as a social issue)
11. How to Behave (as a social issue)
12. Immigration / Visas / Citizenship
13. Job marker (as social issue)
14. Martial Arts & Owning Guns (Social issue)
15. Media (as a social issue)
16. Names (of people)
17. Others' Opinions
18. Owning pets, animal rights & vegetarianism / veganism
19. Parenting (as a social issue)
20. Patent Trolls / Intellectual Property / Solving the Patent Problem
21. Policing
22. Politics & Government (as a social issue)
23. Public Health
24. Race / Racism
25. Religions / Atheism / Agnosticism
26. Romantic / Intimate Relationships: Dating / Courtship / Marriage / Sex / Sexuality
27. Science (as an institution)
28. Social justice topics (race, gender, etc.)
29. Subcultures
30. Suicide
31. War / Genocide / The Military
32. Work (as a social issue)
33. ….

Resources for topics

https://nathanwailes.atlassian.net/wiki/spaces/MTOVT/pages/1409198/Topics+of+Social+Importance

https://ewritingservice.com/blog/social-issues-topics/

https://edubirdie.com/blog/social-issues-topics

https://www.aresearchguide.com/40-social-research-paper-topics.html

FEEL FREE TO ASK CHATGPT

What kind of social issues are most relevant to university students?

University students often encounter a wide range of social issues that impact their academic experience, well-being, and future outlook. Some of the most relevant social issues include:

1. Mental Health

- **Stress, anxiety, and depression** are common due to academic pressures, financial burdens, and uncertainty about the future.
- **Access to mental health resources** on campus is often a concern, as counseling services may be underfunded or have long wait times.

2. Financial Stress and Student Debt

- The rising cost of tuition, books, housing, and other expenses can lead to **financial stress**, forcing many students to work part-time or take on significant debt.
- **Student loan debt** has become a major issue, with many graduates struggling to repay their loans after completing their studies.

3. Diversity, Equity, and Inclusion

- Many students face **discrimination** based on race, ethnicity, gender, or sexual orientation, leading to feelings of exclusion or alienation.
- Issues like **gender inequality**, **LGBTQ+ rights**, and **racial justice** are prominent, and students often advocate for more inclusive policies and curricula on campus.

4. Sexual Harassment and Assault

- **Sexual violence** is a significant issue on many campuses, with concerns about safety, reporting mechanisms, and the effectiveness of university responses.
- There's a growing push for better support for survivors and stricter accountability measures for perpetrators.

5. Climate Change and Environmental Sustainability

- Many students are actively involved in movements to address **climate change**, pushing for more **sustainable practices** on campus and beyond.
- Universities are often challenged to reduce their carbon footprint and integrate sustainability into their operations and teaching.

6. Political Polarization and Free S ech

TOPIC

Start with curiosity - ask why

- What makes you wonder, think?
- What brings you joy, frustration, relief, anger?
- What do you argue about with others?

GROUP FORMATION

YOU (groups of 3)

Split into groups of 3
Spend 5 minutes extracting one topic you care about

Example:

- your data
- … or select a new one, Obesity Crisis, Vaccine Rollout in France, World Climate Goal Progress, …

YOU (groups of 3)

IF you have data already, ask yourselves

- What makes you wonder, think, ask questions?
- What brings you joy, frustration, relief, anger, surprise?
- What do you argue about with others?

ADD YOUR TOPIC TO THIS PAGE

- https://docs.google.com/presentation/d/1I_dGRFHoW05mhmIDJTUiK55sPoH3HY1ePZFKhsDUnBg/edit?usp=sharing
- Copy the 5 template slides for you (see ecampus for this class for the link)
- On the **WHITE** slide add your topic and names

Pick 2 **other group's** slides – on the **green slide** after the topic: (pick the two after yours, skip if they are using the same data)

write down one question about the topic that you would ask yourself (as a student or shareholder)

Think about how the topic brings you joy, frustration, relief, anger, or the kinds of things you have argued about with other people

RESEARCH QUESTIONS

COMPONENTS OF A RESEARCH QUESTION

- Your topic ✓
- Your context (stakeholders) ✓
- Your goal
- Nature of your question

YOUR GOAL

What do you want to achieve?

- Discover
- Describe
- Change too much for a class project but ultimately would be nice
- Explore
- Explain

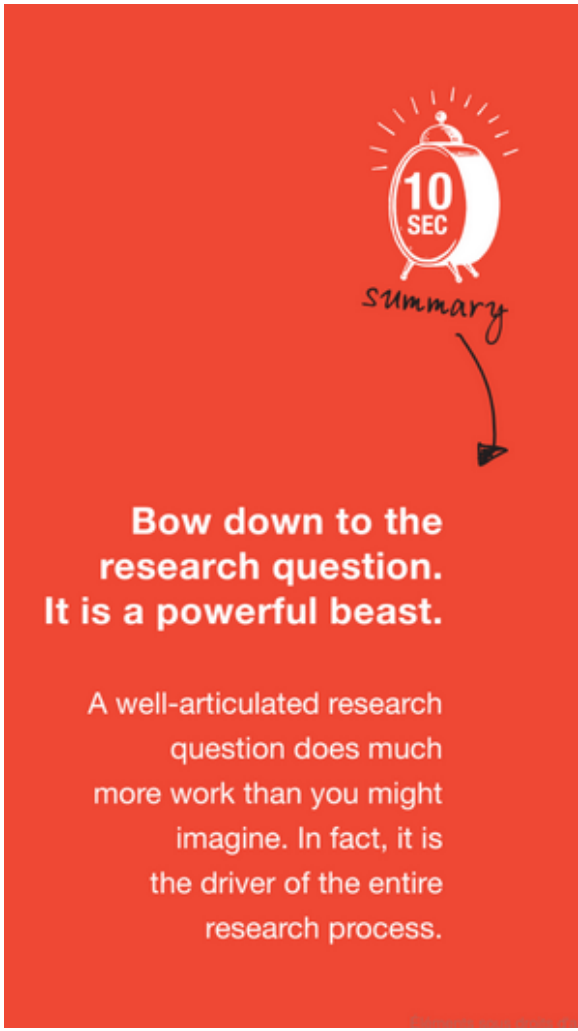
RESEARCH QUESTIONS

- Should be focused on the topic you chose
- Should be highly specific & clear
- Interesting and important
  - It hasn’t already been covered by hundreds of other visualizations on the web
- Appropriately complex

GOOD RESEARCH QUESTIONS

Define

- The topic
- The nature of your endeavour (discover, explore, explain, describe, compare)
- The questions (what, where, how, who, when, why)
- Constructs and variables (age, gender, education,…)
- Potential relationships between variables



Reference: Research Question Little Quick Fix
•Zina O'Leary - The Australia and New Zealand School of Government

CHOOSING A QUESTION

“How has language evolved over time?”

“What will the weather be like next month?”

“Are the right people seeing my advertisements?”

“What is the current temperature?”

A PROBLEM OF SCALE

CHALLENGING TO FIND DATA

"How has language evolved over time?"

"What will the weather be like next month?"

"Are the right people seeing my advertisements?"

"What is the current temperature?"

NOT AS INTERESTING

RESEARCH QUESTIONS

Topic: Obesity crisis:

- How obese are people around the world?
- Are both genders equally impacted by obesity problems?
- How can we solve the obesity crisis?
- What is the obesity crisis?
- What is the difference in the obesity of men and women for different countries around the world?

Which question is the odd one out?

YOUR TURN

Map out the factors that impact your topic on the **purple slide**

Example: Obesity crisis

Which factors impact the extent of the obesity crisis?

https://www.youtube.com/watch?v=KhgxuNvbNrA&t

YOUR TURN

Narrow down your topic based on what info in the green and purple slide and add them the **blue slide**

Example: Obesity crisis

Focus on students and social aspects of obesity

https://www.youtube.com/watch?v=KhgxuNvbNrA&t

YOUR TURN

Go to your own topic and the **yellow slide**

Check out how you have narrowed the topic so far

Write down one question. Think about the **what, who, why, where, when, or how**

YOUR TURN

On your topic. Bold the variables that will help you to conduct a data analysis.

Obesity crisis:

In the USA, how are food prices on college campuses related to the nutritional value of the choices students can make for lunch?

Hint

This work is the bases of part 1 of Assignment 1

(Part 2 is looking for a dataset)

CHECK YOUR QUESTION

IS THE QUESTION RIGHT FOR YOU?

- Can you work on this for the next few weeks?
- Is this question too close to home?
  - Do you have any existing biases that will influence your work?
  - Very personally relevant topics (mental health, diseases, drug use,…) can be healing but also disturbing!

IS THE QUESTION RIGHT
FOR YOUR STAKEHOLDERS?

Is this question significant to a population ?

- Can you say why?
- Can you say what people can do with what they will learn?
- How can your question make a difference?

IS THE QUESTION RIGHT FOR CLASS?

Is this question doable?

- Do you think you can find data on it?
  (questions that require new data collection, ie datasets that do not exist, are not doable in class, that's something to do in your Master or PhD project)
- Do you have the expertise to analyze this topic?
- (Would this question receive ethics clearance?)
- Is this answerable in 1-2 months?

IS THE QUESTION WELL ARTICULATED?

Look at each term in your question

- Make sure it's clear and unambiguous
- Clarify your question if needed

Note: research questions need 3-4 iterations minimum

Spend 5 minutes to rethink your question and improve it

LETS TALK ABOUT EACH QUESTION

(in class if there is time…)
This is your Homework due on **Thursday 19/03** (yes tomorrow) !!!

Share with us a research question and a dataset you found